

Gayathri Emuru

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Open to Relocation | Sponsorship Not Required

Summary

Data scientist with 4 years shipping production ML systems and data pipelines at scale, across fintech, enterprise, research.

- Built infrastructure serving 10M plus users and 700M plus monthly API requests at 99.9% accuracy
- Strong Python, SQL, and AWS cloud, comfortable owning the full lifecycle from ingestion to deployment
- [AWS Certified Machine Learning Engineer Associate](#)
- Co-author of two peer reviewed publications in Nature Portfolio journal

Work Experience

Data Scientist | University of Delaware

Aug 2024 – Present

- Built end to end **ETL and ML pipeline** for a 35,000 plus document multilingual corpus, hitting 83% topic accuracy (sentence transformers, BERTopic, HDBSCAN, OCR, Parquet, joblib, web scraping). Published on [Harvard Dataverse](#) and [Nature Portfolio journal](#).
- Engineered the **data engineering** platform behind a second [Nature Portfolio dataset](#): 100% extraction accuracy, 89.7% geolocation precision, sub 1 km median error across 31,000 plus records (Python, OCR, SQL, FAISS, Bayesian modeling, Nominatim to ArcGIS, Power BI).
- Designed a [RAG extraction system](#) on GPT 5.5 and Opus 4.1 with hybrid retrieval: 87.4% precision on held out tests, validated via A/B benchmarking and 95% bootstrap confidence intervals (FAISS, BM25, RRF).
- Architected **two industry sponsored Python ML pipelines**: a ToF SIMS hyperspectral analysis pipeline at 65,536 pixel scale (PCA, UMAP, SOM), and a 15 class PyTorch ensemble of ResNet50, DenseNet121, and EfficientNetBo hitting 99% accuracy with Grad CAM interpretability (A100 accelerated).

Data Scientist | Nagarro

July 2021 – Aug 2024

- Productionized an **XGBoost risk-classification model** on AWS SageMaker that flagged stocks under Additional Surveillance Measures with 83% accuracy, strengthening the DVM score powering trading decisions for 18 B2B stockbroker clients, directly integrated into client-facing risk dashboards.
- **Built and scaled production REST APIs** for a stock analytics platform serving **700M+ monthly requests**, designing 186 endpoints, real-time market data feeds, clustering-based portfolio segmentation, and **interactive BI dashboards** that drove a **37% increase in monthly active users**.
- Engineered a high-accuracy payments module for Africell Angola (10M users) using Python, NumPy, and Pandas for transaction data pipelines; achieved **99.9% transaction accuracy** and **30% faster settlement speeds** across a financial system operating in a low-connectivity emerging market.

Skills

- **Programming Languages:** Python, SQL, C.
- **Data Analysis & Machine Learning:** Pandas, NumPy, EDA - Data visualization (Matplotlib, Seaborn, Power BI, Tableau), Classical Machine Learning, Deep Learning, NLP, Computer Vision, Statistical methods, XGBoost, LightGBM, AWS, System Design, Scikit-Learn, Generative Models (GMM, GAN's), Image Recognition, Forecasting, Sentiment Analytics, Optimization Techniques, Feature Engineering, Generative AI, LLM.
- **Data Science Processes:** A/B testing, ETL, Data Cleaning, Time Series, Recommender Systems.
- **Software & Tools:** Django (Full Stack), MySQL, Git, GitHub Actions, CI/CD Pipelines, JIRA, REST APIs, Beautiful Soup, Big Data (PySpark), Excel, Linux, TensorFlow, PyTorch, MLFlow, Streamlit, Jupyter Notebooks, R Studio.

Education

University of Delaware
MTIET (A.P - India)

Master of Science in Data Science
Bachelor of Technology(B. Tech) - Computer Science

CGPA: 4/4
GPA:8.5/10

Projects [[GitHub](#)]

- **Generate Anime Images using GAN's: Applied Generative AI(GenAI) - GANs** to create high fidelity Anime images, optimizing model performance during training.
- **County-Level Mental Health Prediction:** Built an end-to-end Python pipeline by integrating multi-source U.S. public health and socioeconomic datasets across 3,100+ counties and 30+ variables, performing feature engineering, imputation, EDA, and visualization, and training **Random Forest and Decision Tree models that achieved $R^2 = 0.71$, $F1 = 0.90$, and $AUC = 0.77$.**
- **Signature Verification:** Built a **Siamese CNN-based Signature Verification** system using Computer Vision, achieving 73% accuracy in distinguishing genuine from forged signatures — directly applicable to fraud detection and identity verification pipelines.
- **Educational Project Approval Prediction** for DonorsChoose.org: Predicted proposal approvals using NLP Techniques and Machine Learning Algorithms achieving a test accuracy of 0.66.